



Thermal-solutal-fluid flow of channel segregation during directional solidification of single-crystal nickel-based superalloys

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ABSTRACT

Channel segregation – known as freckle – is one of the most complex defects which severely limit mechanical properties of investment cast single-crystal turbine blades. The freckle phenomenon is induced by the thermal-solutal-fluid flow instability during solidification process; it is therefore important to gain better understanding of the fluid flow, heat transfer and species transport during the investment casting process. In this work, a three-dimensional Eulerian two-phase model is developed to investigate chimney behaviours and the evolution of channel segregation in the directional solidification of nickel-based single crystal superalloy CMSX-4. The results reveal that the channels are merely distributed on the surface of ingot backed up with targeted experiments, and the morphology of the predicted channel segregation is also consistent with the observed freckle chains. By ignoring lateral heat flux, deviation in the prediction of channel segregation using Rayleigh number criterion is anticipated, because not all the chimneys can be developed into channel segregation under the significant perturbation. In the mushy zone, the perturbation of the melt convection from the horizontal direction to the casting direction leads to the extinction of existing chimneys, thereby exacerbating channel segregation. Once the stable channel segregation is formed, it will be developed steadily under further solute enrichment caused by thermal-solutal convection. Channel segregations is difficult to develop continuously in the centre of the representative rod, while channels near the lateral walls are more likely to develop stable flow along the casting direction. It is suggested that the weaker lateral heat flux can suppress the formation of channel segregation, providing an effective process design criterion for the freckle mitigation during the single crystal casting.

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1. Introduction

Single-crystal turbine blade is considered to be one of the most important components in the aerospace engines. During manufacture of the turbine blade by investment casting, freckle – set of equiaxed grains with no specific orientation distributed in chains – can be found commonly [1]. In the single crystal blade, the grain boundary has been removed to allow only one crystal grain to grow into the entire casting. The occurrence of freckle defect is detrimental to the integrity of the single crystal component in that the grain boundaries between the freckle equiaxed grains degrade the mechanical properties and reduce the stress-rupture life under high-temperature and high-pressure conditions [2,3].

Mechanisms of freckle formation have been reported widely in literature. Hellawell et al. [4,5] conducted visualization exper-

iments, using a transparent $\text{NH}_4\text{Cl-H}_2\text{O}$ system to study the formation of freckle during directional solidification. They observed well-developed chimneys and concluded that free fragments settled in the channels and grew into freckles. It should be noted that chimneys are solute enriched flow, and channel segregation is the uneven solute distribution in solidified metal. Laboratory X-ray radiography was used to observe the directional solidification of Ga-In alloys [6,7], and it was confirmed that the dendritic fragments caused by remelting tended to stay in the segregation channel. These findings explained why freckle was concentrated on the segregation channel of superalloy castings, and confirmed channel segregation was the essential condition for freckle formation. In a dendritic-scale simulation, Yuan and Lee (2012) [8] and Karagadde et al. (2014) [9] used the coupled cellular automaton and finite difference method to reveal the freckle initiation. They reported that the solute enrichment between dendrite arms caused by different dendritic growth rates was the origin of channel segregation formation.

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Nomenclature

c_0	Initial concentration wt.%
c_l, c_s	Species concentration wt.%
c_l^*, c_s^*	Interface equilibrium solute concentration wt.%
C_{ls}^M	Species exchange $\text{kg m}^{-3} \text{ s}^{-1}$
c_{mix}	Mixed concentration wt.%
c_p^l, c_p^s	Specific heat $\text{J kg}^{-1} \text{ K}^{-1}$
D_l	Solute diffusion coefficient $\text{m}^2 \text{ s}^{-1}$
d_c	Diameter of primary dendrite arm trunk m
f_l, f_s	Volume fraction
$\bar{g}$	Gravity m s^{-2}
H^*	Volume heat transfer coefficient $\text{W m}^{-3} \text{ K}^{-1}$
h	Heat transfer coefficient $\text{W m}^{-2} \text{ K}^{-1}$
h_l, h_s	Enthalpy J kg^{-1}
h_l^{ref}, h_s^{ref}	Enthalpy at reference temperature J kg^{-1}
Δh_f	Heat of fusion J kg^{-1}
K	Permeability in mushy zone m^2
k	Solute equilibrium partitioning coefficient
k_l, k_s	Thermal conductivity $\text{W m}^{-1} \text{ K}^{-1}$
M_{ls}, M_{lc}	Mass transfer rate $\text{kg m}^{-3} \text{ s}^{-1}$
m	Slope of the liquidus $\text{K (wt.\%)}^{-1}$
p	Pressure N m^{-2}
Q_{ls}^D	Energy exchange rate between phases $\text{J m}^{-3} \text{ s}^{-1}$
Q_l^M, Q_s^M	Energy exchange rate due to phase change $\text{J m}^{-3} \text{ s}^{-1}$
S_A	Surface area concentration m^{-1}
SI	Dimensionless segregation index
R_c	Radius of primary dendrite arm trunk m
R_f	Far field radius of primary dendrite arm trunk m
T, T_l, T_s	Temperature K
T_{ref}	Reference temperature K
t	Time s
$\vec{U}_{ls}^D$	Momentum exchange due to drag force $\text{kg m}^{-2} \text{ s}^{-2}$
$\vec{U}_{ls}^M$	Momentum exchange rate due to phase change $\text{kg m}^{-2} \text{ s}^{-2}$
$\vec{u}_l$	Velocity m s^{-1}
v_{Rc}	Dendritic growth speed along radius direction m s^{-1}
β_c	Solutal expansion coefficient $\text{wt.\%}^{-1}$
β_T	Thermal expansion coefficient K^{-1}
Φ_{imp}	dendrite growing surface impingement
λ_1	Primary dendrite arm spacing m
λ_2	Secondary dendrite arm spacing m
ρ_l, ρ_s	Density kg m^{-3}
μ_l	Viscosity $\text{kg m}^{-1} \text{ s}^{-1}$
$\bar{\epsilon}$	Stress-strain tensors $\text{kg m}^{-1} \text{ s}^{-2}$
Subscripts	
l, s	liquid and solid phase

Macroscopic mathematical model can also be used to predict freckle formation. Using the continuum mixture model established by Bennon and Incropera [10], Felicelli et al. [11,12] simulated heat transfer, species transfer and fluid flow, and predicted channel segregation on macro-scale. Based on the physics of Rayleigh-Bernard flow instabilities, Worster [13] attributed the formation of chimneys to the thermal-solutal convection, and proposed the Rayleigh number in the mushy zone as a criterion to estimate the possibility of freckle formation in castings. Beckermann et al. [14,15] developed a two-phase volume-averaged mathematical model to simulate the formation of freckles, and defined Rayleigh number as the function of thermal conductivity, viscosity, mean permeability

of mushy zone, the height of mushy zone and the density difference between the top and bottom of the mushy zone. According to the simulation results and experimental data [16,17], they gave the critical Rayleigh number to predict freckles in superalloys CMSX-2 castings.

However, the Rayleigh number criterion could not work as well as expected in application, because the lateral heat flux in directional solidification was ignored, and researchers had to modify the critical Rayleigh number under specific conditions [18–21]. Although the thermal conductivity of the shell mould used in nickel-based superalloys investment casting is poor, there is still a strong lateral heat flux during the withdrawal process [22]. In the study on the directional solidification of Ga-In alloys, Saad et al. [23] shown the significant impact of lateral temperature on channel formation. Additionally, the previous investigations on freckles in the other casting processes proved the sensitivity of freckles to the temperature field [24–28]. Furthermore, Ma et al. [29,30] also found that the freckles only appeared on the surface of the superalloy specimen during directional solidification process. On the scale of microstructure, Reinhart et al. [31] also observed the same specific distribution of channel in the in-situ and real-time X-radiography. This was completely different from the theoretical analysis that freckles tended to appear in the region with poor heat dissipation [1,32]. This was also different from the previous simulation results that channel segregation was formed in both centre and edge [12,14]. The understanding of specific distribution of freckles is crucial to the reduce or even eliminate the defect in superalloy castings, but obviously it has not been well explained yet.

In this work, we use the previously developed Eulerian two-phase model to simulate the three-dimensional channel segregation in the directional solidification of nickel-based superalloys CMSX-4. The lateral heat flux in Bridgman method is considered. The simulation result is validated by comparing the predicted channel segregation with the freckles observed in experimental results. The mechanism of the specific distribution of channel segregation is discussed, and the formation and disappearance of chimneys and channels are presented and rationalised. The sensitivity of channel segregation to cooling condition and geometric characteristics are also studied to further analyse the freckles in the actual manufacturing of nickel-based single-crystal superalloy turbine blade.

2. Thermal-chemical-fluid flow

To rationalise the thermal-chemical-fluid flow during directional solidification of single crystal superalloys, an Eulerian two-phase model adopted from Li et al. [26,33] is employed to describe the fluid flow, heat and mass transfer during the process. The governing equations, source terms and auxiliary equations are listed in Table 1.

Two phases of liquid metal and solid dendrite trunks are defined in this model. The dendrite trunks are approximated to be immobile staggered cylinders extending straight up from the bottom on sub-grid scale. Thermodynamic equilibrium condition is assumed at the solid/liquid interface, dictating the interfacial solute concentration in liquid and solid (c_l^*, c_s^*). The growth of dendrite trunks is governed by interfacial solute diffusion, and the driving force is the difference in solutal concentration. Mass transfer rate between solid and liquid, M_{ls} , is calculated with the radial growing velocity of the dendrite trunks. Under the two-phase framework, the local solute concentration is characterised by the volume averaged concentration of the two phases, c_{mix} . Segregation index (SI) is defined as

$$SI = (c_{mix} - c_0)/c_0 \quad (23)$$

Table 1
List of governing equations, source terms and auxiliary equations [26,33,34].

Governing equations	
Phase transformation	Mass conservation
	$\frac{\partial}{\partial t}(\rho_l f_l) + \nabla \cdot (\rho_l f_l \bar{u}_l) = -M_{ls}$ (1)
	$\frac{\partial}{\partial t}(\rho_s f_s) = M_{ls}$ (2)
	$f_l + f_s = 1$ (3)
	Mass transfer rate
	$M_{ls} = v_{R_c} \cdot S_A \cdot \rho_s \cdot \Phi_{imp}$ (4)
	Growing velocity of primary dendrite arm trunks
	$v_{R_c} = \frac{dR_c}{dt} = \frac{D_l}{R_c} \cdot \frac{c_l^* - c_l}{c_l^* - c_s} \cdot \ln^{-1}\left(\frac{R_c}{R_c^*}\right)$ (5)
	Diameter of primary dendrite arm trunks
	$R_c = \frac{1}{2} \lambda_1 \cdot \sqrt{\frac{\sqrt{12} f_s}{\pi}}$ (6)
Fluid flow	Far field radius of primary dendrite arm trunks
	$R_f = \frac{1}{\sqrt{3}} \lambda_1$ (7)
	Dendritic structure surface concentration
	$S_A = \frac{4\pi R_c}{\sqrt{3} \lambda_1^2}$ (8)
	Growing surface impingement
	$\Phi_{imp} = \begin{cases} 1 & R_c \leq \lambda_1/2 \\ 2\sqrt{3} f_l / (2\sqrt{3} - \pi) & R_c > \lambda_1/2 \end{cases}$ (9)
	Momentum equation
	$\frac{\partial(\rho_l f_l \bar{u}_l)}{\partial t} + \nabla \cdot (\rho_l f_l \bar{u}_l \otimes \bar{u}_l) = -f_l \nabla p + \nabla \cdot \bar{\tau}_l + \rho_l f_l \bar{g}_l - \bar{U}_{ls}^M - \bar{U}_{ls}^d$ (10)
	Thermal-solutal buoyance
	$\bar{g}_l = [\beta_T (T_{ref} - T) + \beta_C (c_0 - c)] \bar{g}$ (11)
Heat transfer	Momentum exchange term
	$\bar{U}_{ls}^M = \begin{cases} \bar{u}_l \cdot M_{ls} & \text{solidification} \\ 0 & \text{melting} \end{cases}$ (12)
	Drag force between liquid and solid phase
	$\bar{U}_{ls}^d = \frac{\mu_l f_l^2}{K} \bar{u}_l$ (13)
	Mushy zone permeability
	$K = 6 \times 10^{-4} \cdot \lambda^2 \cdot \frac{f_l^3}{(1-f_l)^2}$ (14)
	Energy conservation
	$\frac{\partial(\rho_l f_l h_l)}{\partial t} + \nabla \cdot (\rho_l f_l \bar{u}_l h_l) = \nabla \cdot (\rho_l f_l k_l \nabla T) + Q_{ls}^M - Q_{ls}^H$ (15)
	$\frac{\partial(\rho_s f_s h_s)}{\partial t} = \nabla \cdot (\rho_s f_s k_s \nabla T) + Q_{ls}^M + Q_{ls}^H$ (16)
	Enthalpy transfer
	$Q_{ls}^M = \begin{cases} (\Delta h_f \cdot f_l - h_l) \cdot M_{ls} & \text{solidification} \\ (\Delta h_f \cdot f_s + h_s) \cdot M_{ls} & \text{melting} \end{cases}$ (17)
Solute transport	Latent heat
	$Q_{ls}^H = H^* \cdot (T_f - T_c)$ (18)
	Species conservation
	$\frac{\partial(\rho_l f_l c_l)}{\partial t} + \nabla \cdot (\rho_l f_l \bar{u}_l c_l) = -C_{ls}^M$ (19)
	$\frac{\partial(\rho_s f_s c_s)}{\partial t} = C_{ls}^M$ (20)
	Solute redistribution
	$C_{ls}^M = \begin{cases} k \cdot c_l^* \cdot M_{ls} & \text{solidification} \\ c_s \cdot M_{ls} & \text{melting} \end{cases}$ (21)
	Mixed concentration
	$c_{mix} = (\rho_l f_l c_l + \rho_s f_s c_s) / (\rho_l f_l + \rho_s f_s)$ (22)

to estimate the deviation between local concentration and initial concentration.

We treat the nickel-based superalloys as eight Ni-X binary systems ($X = \text{Ta, Cr, Al, Co, W, Re, Ti, Mo}$). The phase diagram of each Ni-X binary system is linearized with a constant partition coefficient and a constant liquidus slope. Then the multicomponent superalloys are simplified as one equivalent Ni-X binary system. The corresponding equivalent mass fraction $\bar{c}_0$, partition coefficient $\bar{k}$, slope of liquidus $\bar{m}$ and solutal expansion coefficient β_c can be calculated with the following equations:

$$\begin{cases} \bar{c}_0 = \sum_{i=1}^{N-1} c_{0i} \\ \bar{c}_0 \cdot \bar{m} = \sum_{i=1}^{N-1} m_i c_{0i} \\ \bar{c}_0 \cdot \bar{m} / \bar{k} = \sum_{i=1}^{N-1} m_i c_{0i} / k_i \\ \bar{c}_0 \cdot (1 - \bar{k}) \cdot \beta_c = \sum_{i=1}^{N-1} c_i (1 - k_i) \beta_{c_i} \end{cases} \quad (24)$$

Natural convection is modelled with Boussinesq approximation, where the change in density of molten metal with temperature and solute concentration is treated as a body force. The mushy zone, that is the mixture of dendrite arms and the residual liquid metal, is treated as a porous media [35–37]. Solid volume fraction is employed to describe the solidification procedure, and the solidification process ends only if the solid volume fraction reach unity, when no residual liquid left in the interdendritic region [10,38].

The interdendritic flow resistance in mushy zone, $\bar{U}_{ls}^D$, is described with Darcy's law, where the drag force between phases is a function of permeability [39]. Blake-Kozeny equation is employed to calculate the permeability, where the characteristic length is set as dendrite arm spacing [40]. In single-crystal castings, the primary dendrite arms grow along the casting direction $\langle 001 \rangle$. Thus, the characteristic length for the permeability parallel to $\langle 001 \rangle$ is primary dendrite arm spacing, and the one perpendicular to $\langle 001 \rangle$ is secondary dendrite arm spacing. Due to the local thermal equilibrium, a volumetric heat transfer rate between the two phases of $10^9 \text{ W m}^{-2} \text{ K}^{-1}$ is defined to level out the temperature difference.

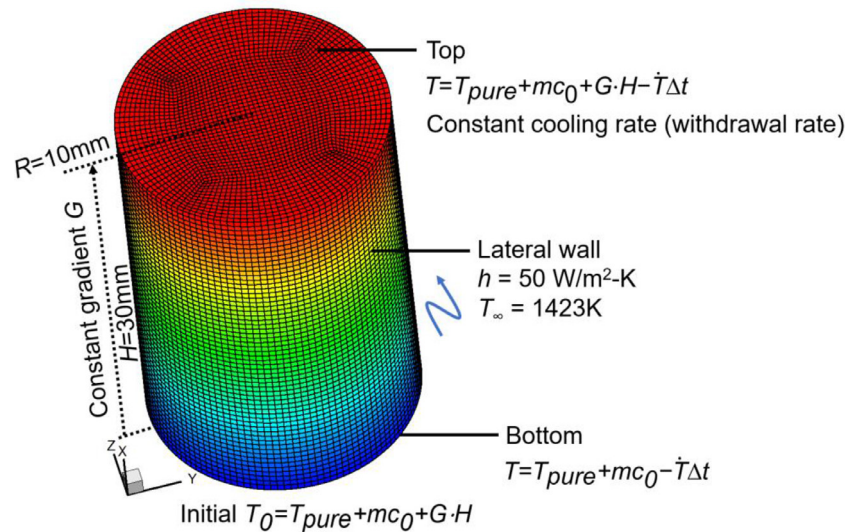


Fig. 1. Geometry and mesh of the superalloy cylinder specimen. Boundary conditions and initial conditions in the benchmark case.

Table 2
Thermophysical properties of CMSX-4.

Properties	Value	Reference
Density (kg/m ³)	7646	[47]
Thermal conductivity of liquid (W/m·K)	33.5	[47]
Thermal conductivity of solid (W/m·K)	24.6	[47,48]
Specific heat (J/kg·K)	575	[47]
Latent heat (J/kg)	240,000	[48]
Diffusion coefficient in liquid (m ² /s)	3.6×10^{-9}	[49]
Thermal expansion coefficient (K ⁻¹)	1.157×10^{-4}	[12]
Viscosity (kg/m·s)	4.90×10^{-3}	[12]
Pure melting point (K)	1726	[50]
Eutectic temperature (K)	1593	[47]

3. Simulation parameters

The benchmark cylinder specimen with a diameter of 20 mm and a height of 30 mm is used in our simulation. The previous investigations [25,26,41,42] have shown the mesh sensitivity of channel segregation and concluded that the grid size should be comparable to the primary dendritic arm spacing to capture the details of channel segregation. Thus, the average grid size in the simulation is set to 500 μm , which is equal to the primary dendrite arm spacing, so as to accurately capture the details of channel segregation. Fig. 1 shows the mesh, where the computational domain is divided into 153,600 hexahedral control volumes. The casting direction is along X-axis.

The CMSX-4 nickel-based superalloy is used in this work. The thermophysical properties of CMSX-4 are listed in Table 2, and the nominal composition and phase-diagram parameters are listed

in Table 3. The constant temperature gradient of 2000 K/m along casting direction and constant cooling rate of 0.1 K/s are set, corresponding to the withdrawal rate of 50 $\mu\text{m/s}$ in Bridgman method. The convective heat transfer condition is applied to the lateral wall, where the heat transfer coefficient is 50 W/m²·K and the freestream temperature is 1423 K. According to the experimental data and simulation [43,44], the primary dendrite arm spacing is set to 500 μm , and the secondary dendrite arm spacing is 100 μm [45,46]. All the boundaries of the computational domain are no-slip boundaries.

Based on finite volume method, phase-coupled SIMPLE method is used to solve the transport equations listed in Table 1. The auxiliary variables, exchange terms and source terms are updated before the iterations in a timestep. Up to 60 iterations are performed in each 0.1 s timestep to reduce the normalized residuals. The convergence criterions for continuity equations, momentum equations and solute equations are all 10^{-3} , while it is 10^{-6} for enthalpy equations.

4. Results

4.1. Verification of the model

Ma et al. [29,30] found that the freckle only appeared on the surface of the superalloy specimen during directional solidification process. To the best of our knowledge, no model has, thus far, predicted the surface channel segregation. The predicted channels in the previous simulation results are mainly distributed in the centre [12,14]. In our simulation result, channel segregation is only formed on the surface of specimen, extending from bottom

Table 3
Composition, partition coefficient, slope of liquidus and solutal expansion coefficient of CMSX-4.

Element	Mass fraction wt.%	Slope of liquidus K/wt.%	Partition coefficient	Solutal expansion coefficient (wt.%) ⁻¹
Ta	6.5	-5	0.78	-4.60×10^{-3}
Cr	6.5	-2.15	0.95	1.90×10^{-3}
Al	5.6	-5.35	0.91	2.50×10^{-2}
Co	9	0.01	1.08	1.13×10^{-4}
W	6	2.45	1.28	-5.30×10^{-3}
Re	3	5.45	1.73	-5.46×10^{-3}
Ti	1	-16.3	0.71	8.20×10^{-3}
Mo	0.6	-5.72	1.06	-1.80×10^{-3}
Reference	[51]	[11,12,43]	[14,51-53]	[14,52]

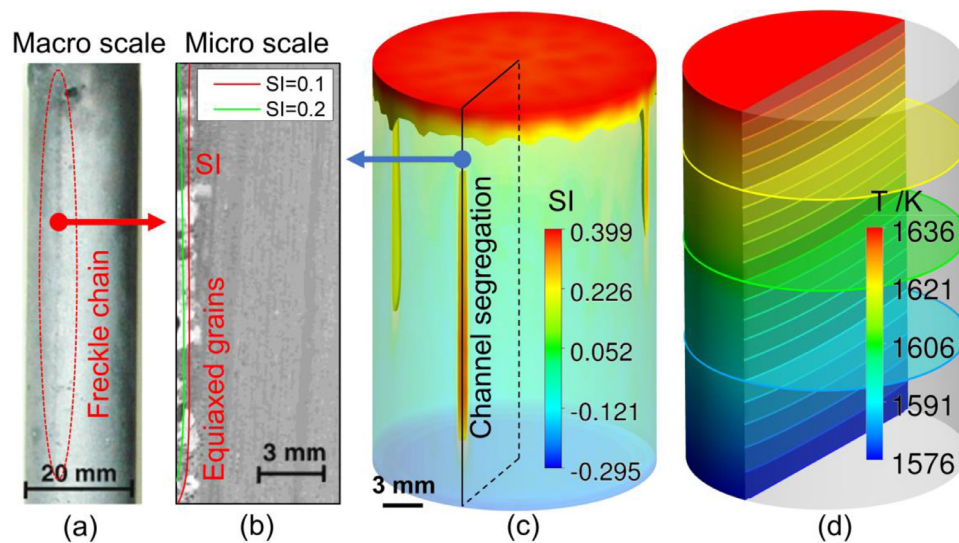


Fig. 2. Comparison between the experimental data and simulation results. (a) Freckle chain in directional solidification ingot of nickel-based superalloy CMSX-4 observed by Ma and Bührig-Polaczek (2014) [30]. (b) Comparison between freckle layer depth [30] and channel width. Iso-surfaces of $SI=0.1$ and $SI=0.2$ are used to mark the boundary of channel. (c) Simulated distribution and morphology of channel segregation. The isosurface of $SI=0.2$ is used to represent the channels. (d) Temperature distribution at the solidification time of 850 s.

to top, as shown in Fig. 2c. The distribution and morphology of channels are consistent with the freckle chains observed in the experiments conducted by Ma et al. [29,30], as shown in Fig. 2a. The comparison of observed freckle layer depth and channel width is given in Fig. 2b. Since it is difficult to quantitatively associate segregation index with the formation of freckles, two isosurfaces of $SI=0.1$ and $SI=0.2$ are given to represent channel boundary. In the previous experimental results [29,30,54], the depth of freckle layer is mostly between one and two times primary dendrite arm spacing, and the average distance between the calculated isosurface of $SI=0.1$ and wall is 579 μm . Fig. 2d shows the predicted temperature distribution of the cylinder specimen at the solidification time of 850 s. Due to the constant temperature gradient is applied to the computational domain, as it is discussed and shown in Fig. 1, the presented temperature field does not change significantly during the solidification process. The temperature isotherm is concave because of the heat loss through the lateral wall, which is well compared with the previous investigations [47,55,56] on the heat transfer in the actual directional solidification of superalloy castings with Bridgman method. From above, the simulation results are in good agreement with experimental data, so the model can provide reasonable results for further discussion.

4.2. Evolution of chimneys and channel segregation

In order to clearly show the evolution of chimneys, the different SI isosurfaces are used at different times. At 50 s, a strong upward chimney is formed in the centre, which is much stronger than the chimneys in the near-wall region, as shown in Fig. 3a. But more and stronger chimneys are subsequently observed in the near-wall region, and then a channel is formed under one of these chimneys, as shown in Fig. 3b and c. The chimneys keep swinging, while the channel extends straight upwards. As the solidification proceeds, the dominant chimneys are distributed on the surface, while the central chimney dominates flow field in the early stage. When the channel grows longer, the SI in the channels becomes higher than the SI in the chimneys, as shown in Fig. 3d. The solute distribution parallel to casting direction is extremely uneven, and chimney flow becomes much weaker. Interestingly, there are three elongated channels segregation on the surface in the final ingot, while no channel is found in the centre, as shown in Fig. 2c. One extends

longitudinally from the bottom and ends near the top; one extends from the middle to the top, and the last one only appears on the upper part of the ingot.

The evolution of solute distribution on the lateral surface and longitudinal sections is shown in Fig. 4. As shown in Fig. 4a, in the initial stage of directional solidification, short chimneys of different lengths are formed on the surface of specimen. At 200 s, only one channel is formed under only one of these chimneys, as shown in Fig. 4d. Additionally, there are seven chimneys in Fig. 4a and nine chimneys in Fig. 4d, indicating the formation, disappearance, and movement of chimneys in the solidification front. The SI in the channels becomes higher and higher during the solidification procedure, as shown in Fig. 4g and j.

Fig. 4b, e, f and k show the evolution of solute field and flow field on Section 1. In the initial stage of solidification ($t = 50$ s), due to the higher cooling rate, more solute is rejected into melt in the near-wall region, as shown in Fig. 4b. Descend flow driven by thermal buoyancy mixes with the upward flow driven by solutal buoyancy. Under the influence of melt flow in the horizontal direction, the flow field then becomes asymmetric, and the segregation tends to be interrupted and cannot develop along casting direction continuously, as shown in Fig. 4e and h. At 450 s, the flow intensity gets much weaker, and flow field tends to be relatively uniform. No channel segregation can be observed on this section, as shown in Fig. 4k. Fig. 4c, f, i and l demonstrate the evolution of flow field and solute distribution on Section 2 where one of the channels in the final ingot is located. As shown in Fig. 4c, in the initial stage, the ascend flow driven by solutal buoyancy competes with the descend flow driven by thermal buoyancy in the near-wall region, thereby inhibiting the chimney flowing up directly, intensifying the local solute enrichment and promoting the formation of chimneys. The central chimney then disappears, while a chimney is formed near the lateral wall, as shown in Fig. 4f. The channel develops continuously under the stable upward chimney as shown in Fig. 4i and l.

The evolution of segregation index distribution on the cross-section of $X = 20$ mm is shown in Fig. 5. Similar to the results shown in Figs. 3a-c and 5a and b demonstrates the transition of flow field from central chimneys dominating to near-wall chimneys dominating. It should be noticed that the solidification front becomes quite uneven during the moving of chimneys. As shown in

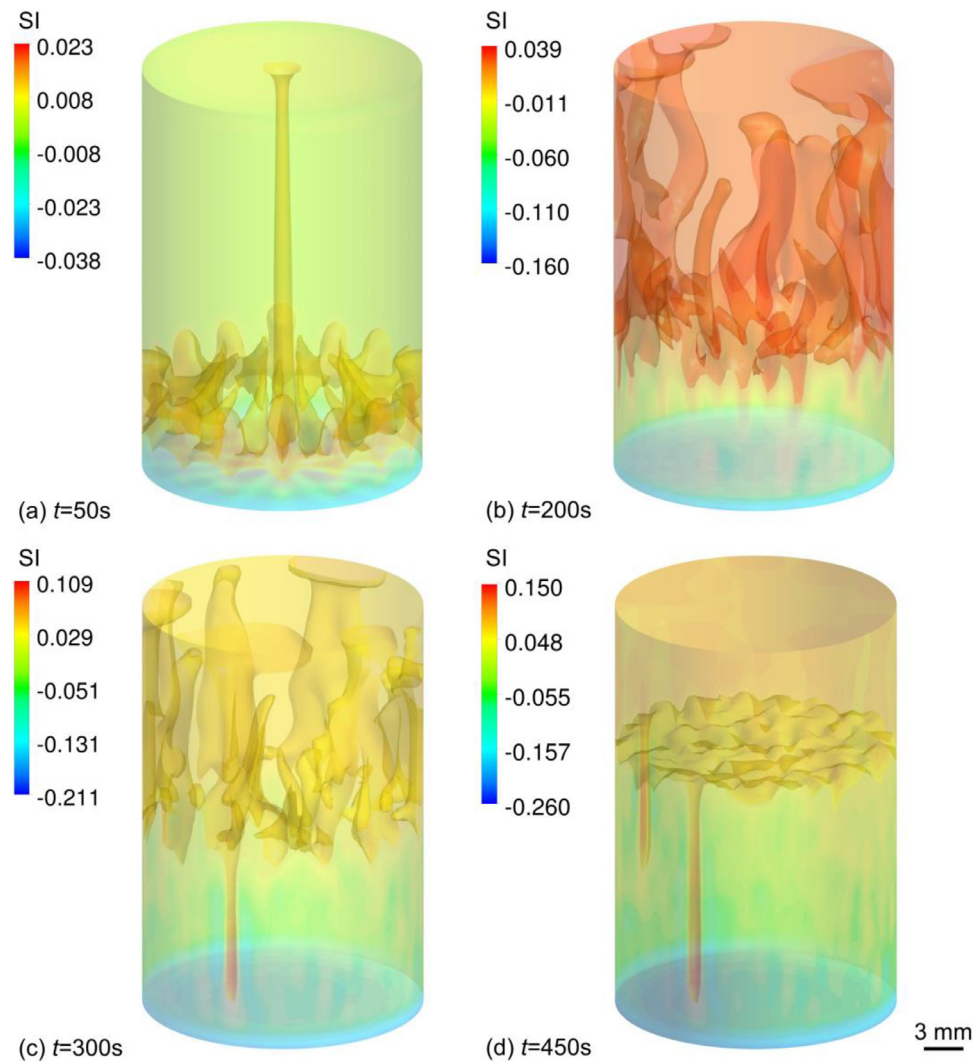


Fig. 3. The evolution of chimneys and channel segregation during the solidification process. Transparent effect is used in these group of sub-figures. Iso-surfaces are used to characterize the chimneys. Iso-surface of $SI=0.007$ is used at (a) $t = 50$ s, iso-surface of $SI=0.026$ is used at (b) $t = 200$ s, iso-surface of $SI=0.045$ is used at (c) $t = 300$ s, iso-surface of $SI=0.06$ is used at (d) $t = 450$ s.

Fig. 5c, when the solidification front approaches the cross-section, many solute enriched chimneys are formed in both the centre and near-wall region. However, when the solidification front advances above the cross-section, only three channels are formed in the near-wall region, while the solute distribution in the centre becomes relatively uniform, as shown in Fig. 5d.

5. Discussion

5.1. Stabilization of channel segregation and disappearance of chimneys

The segregation distribution along one of the channels during solidification process is shown in Fig. 6, which demonstrates the development of channel segregation. Once a channel is formed, solute is further rejected into the channel, resulting in an increase in the segregation index [57,58]. Due to the further solute enrichment, the local solidification procedure is further delayed. The solute concentration gradient along casting direction drives strong upward flow with enriched solute, thereby enhancing the stability of the channel. At the top of the specimen, as the solute concentration in melt gets higher and solute enriched upward flow weakens,

the local constitutional undercooling reduced significantly and the top of the channel becomes more sensitive to the melt flow in bulk region. Thus, the channel tends to be interrupted.

The segregation indexes along Channel 2, Channel 3, Line 4 (on the lateral wall) and centre-line are shown in Fig. 7. The segregation index fluctuates along the centre-line, which indicates that channels cannot develop stably under the strong perturbation caused by melt convection. The segregation pattern along the Line 4 on the lateral wall is similar. The rapid increase of segregation index along casting direction represents a largely accumulation of enriched solute, indicating the developing of channel segregation. Channel 2 develops from the bottom and segregation index increases rapidly along casting direction, while Channel 3 begins to develop continuously at about one-third of height. Comparing the segregation index fluctuations along Line 4, centre-line and Channel 3, it can be inferred that the perturbation near the wall is weaker than the centre. Similar to the evolution of segregation index in Channel 2, when Channel 3 starts to form, the solute is rapidly accumulated, thereby stabilizing the channel. It can be deduced that the perturbation caused by the melt flow in the horizontal direction can hardly interrupt the channels in the developing stage because the further enriched solute at the bottom of

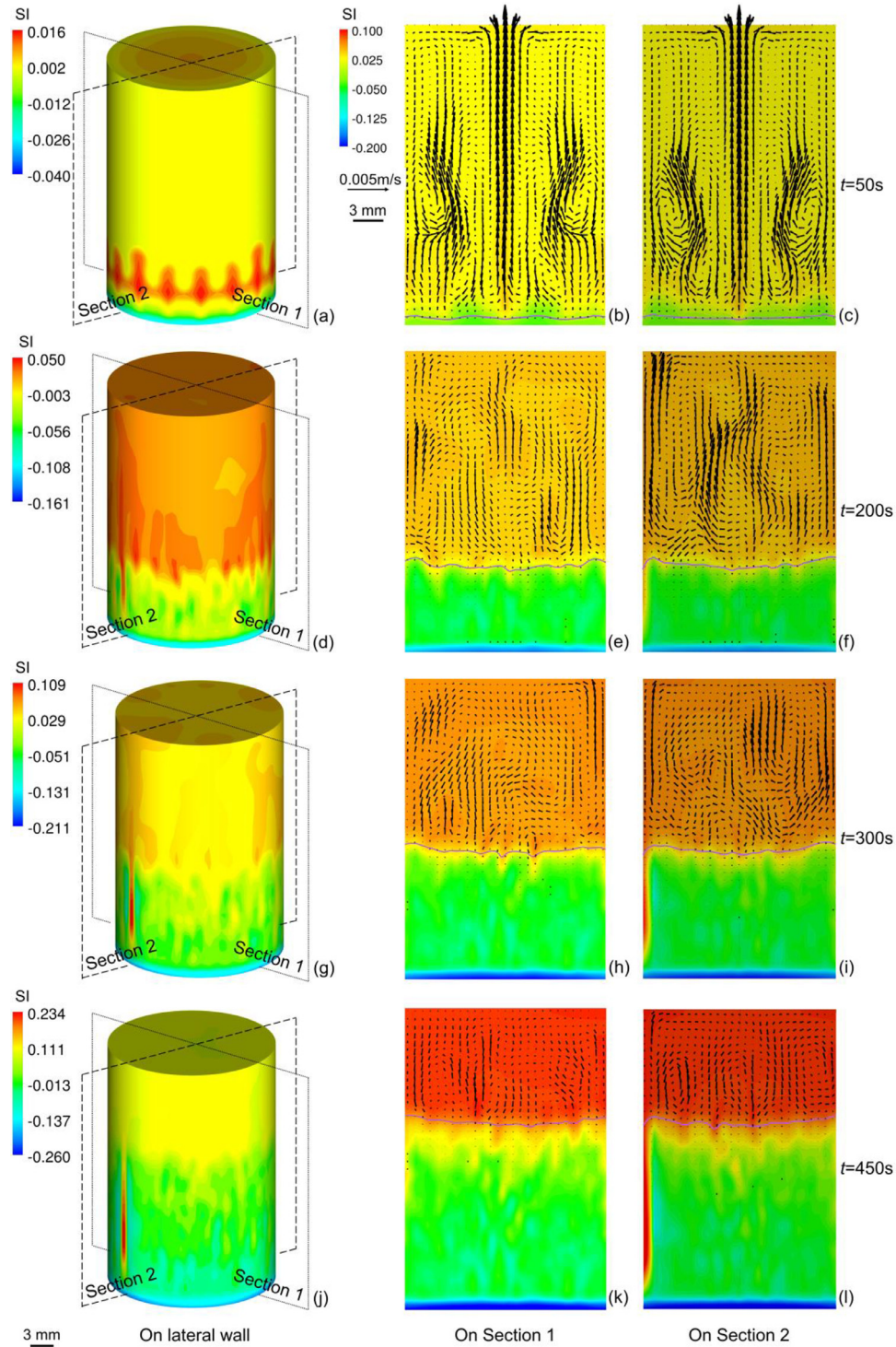


Fig. 4. Segregation index distribution and flow field at a) 50 s, b) 200 s, c) 300 s and d) 450 s on the lateral surface and longitudinal sections. One channel can be observed on [Section 2](#), while no channel is observed on [Section 1](#). Iso-surface of $f_s=0.05$ is used to mark the solidification front (the envelop of dendritic tips [28]).

channel drives stable upward flow. However, as the channel gets longer, the chimney flow becomes much weaker, and it is easier to be influenced by the surrounding melt convection.

According to the previous investigations, the formation of chimney was ascribed to the density inversion in the mushy zone [11,14]. Based on this understanding, Rayleigh number is often employed to estimate the probability of freckles. The driving force for the formation of freckles is described by the competition between the solutal buoyancy caused by solute enrichment and the

flow resistance in the mushy zone. It should also be noticed that what the Rayleigh number actually estimates is the possibility of chimney formation. Considered as a sign of the freckles, chimney was always used to predict freckles in previous studies [7,14,36]. However, the heat flux on the lateral wall was not considered in these investigations. As discussed above, the perturbation induced by radial cooling conditions also has a great impact on the interdendritic flow and solute distribution. It is important to note that chimneys seem quite unstable as demonstrated in [Figs. 3 and 4](#).

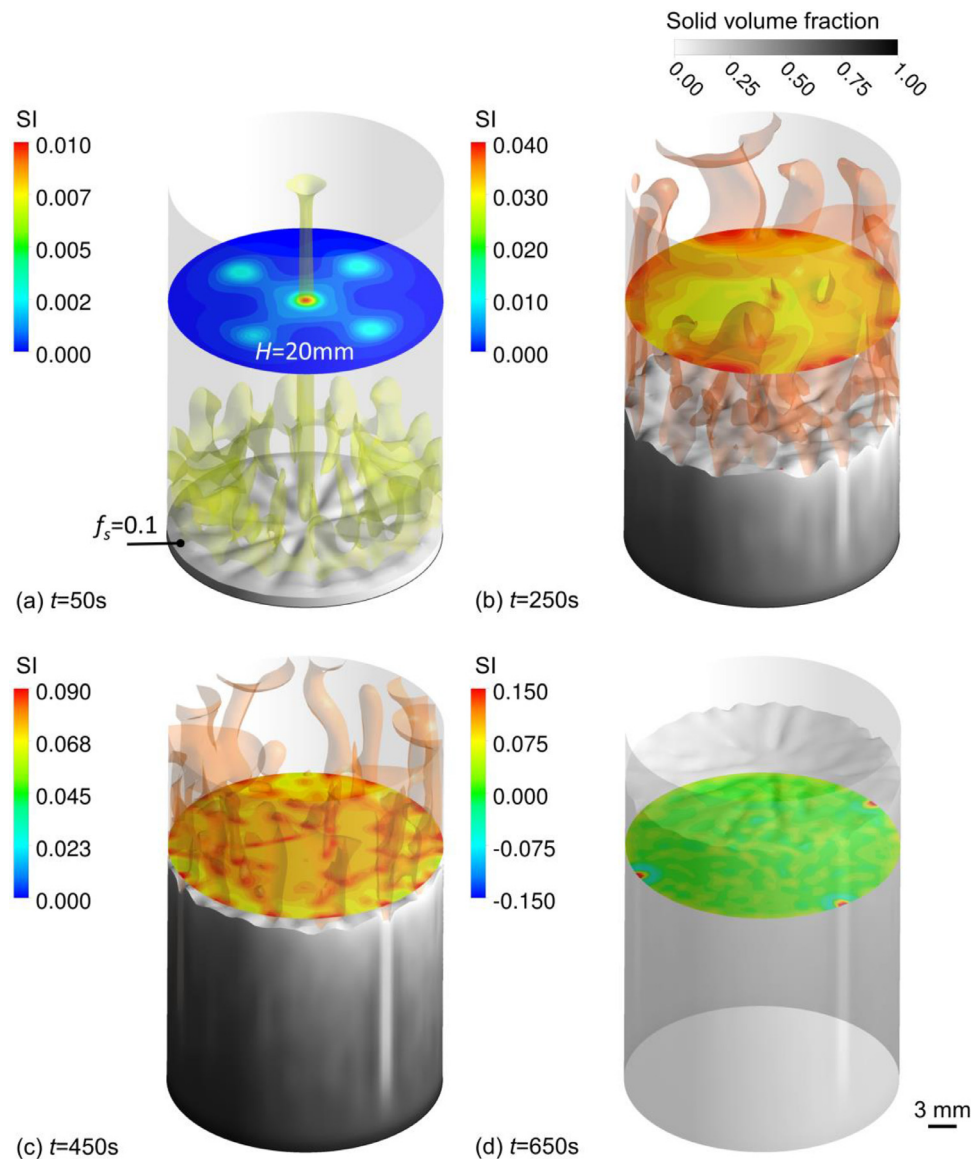


Fig. 5. Evolution of segregation index distribution on the cross-section of $H = 20$ mm. The isosurface of $f_s=0.1$ is used to represent the solidification front. The control volumes where $f_s \geq 0.1$ is coloured with f_s in grey scale. The iso-surfaces of SI are used to show the chimney flow above the solidification front. Iso-surface of $SI=0.007$ is used at (a) $t = 50$ s, iso-surface of $SI=0.035$ is used at (b) $t = 250$ s, iso-surface of $SI=0.0825$ is used at (c) $t = 450$ s, and transparent effect is employed in the solidified region for a better view of the segregation index distribution in the cross-section at (d) $t = 650$ s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Not all the chimneys can be retained and continue to develop until channel segregation is formed underneath. Thus, there is always a deviation between the prediction results based on the Rayleigh number criterion and the experimental statistics. Fig. 8 is a schematic of the interaction amongst solute distribution, melt convection and chimneys flow. The thermal-solutal convection always changes the solute distribution in mushy zone. Under the perturbation induced by melt convection, chimneys move, randomly appear and disappear at the solidification front, as it is demonstrated in Fig. 8a. The upward chimney brings the rejected solute into the channel, resulting in an increase in segregation index and further stabilizing the channels as shown in Fig. 8a. Things are similar in the near-wall region, but the melt convection should be divided into tangential component and normal component to estimate the amplitude of perturbation induced by melt convection, as shown in Fig. 8b.

As discussed above, the melt convection induces perturbation on chimneys along horizontal direction. A quantitative analysis is

performed to further confirm the behaviours of chimneys. Fig. 9 shows the evolution of magnitude of horizontal velocity component and solute distribution along solidification front on longitudinal sections. The peaks of solute concentration curve indicate that the roots of chimneys are located here. On Section 1 at 50 s, there is a difference between magnitudes of velocity component in the horizontal direction on the two sides of the chimney, as shown in Fig. 9a. For the chimney in the centre, the enriched solute in the root will be transferred to the left side and inhibit the advancement of the local solidification front, while the solidification front below the chimney will advance faster with the enriched solute being taken away. Thus, the chimneys in the centre and mid-radius region move to the left. At 200 s, magnitudes of velocity component on both sides of chimneys are higher and still different, which continue to promote the movement of chimneys, as shown in Fig. 9b. Although the solute difference between the peak and alley gets larger, under the perturbation of melt convection, the rejected enriched solute is still transferred along solidifi-

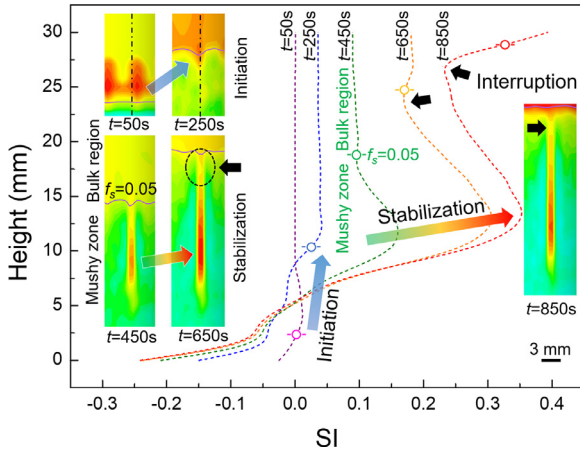


Fig. 6. Evolution of segregation index distribution along the channel during solidification process. The position of $f_s=0.05$ is plotted to distinguish between the bulk region and mushy zone. The contours of segregation index distribution near the channel at the times are also plotted.

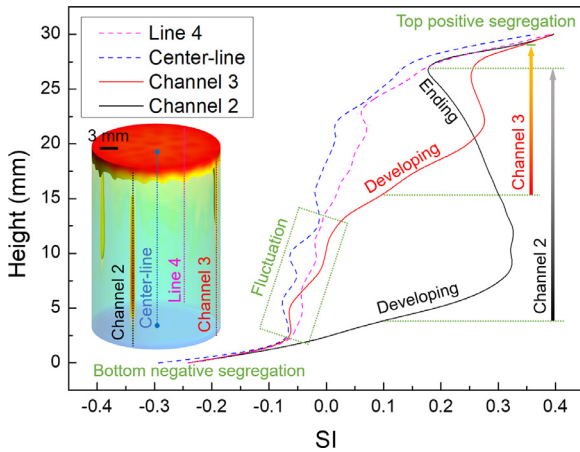


Fig. 7. Segregation index distribution along Line-4, centre-line and Channel 2 and 3 at 850 s. Line 4, Channel 2 and centre-line are in the same plane.

cation front, making the chimneys quite unstable. With the directional solidification, the chimneys keep swinging under the effect of thermal-solutal convection. Thus, no matter the original chimneys and newly forming chimneys, they cannot develop continuously under the perturbation along the solidification front. As a result, channel segregation will not be observed in the centre of the ingot.

The evolution of magnitude of horizontal velocity component and solute distribution along solidification front on Section 2 are demonstrated in Fig. 9d-f. Though the flow fields at 50 s is relatively symmetric, the flow field still becomes asymmetric due to the accumulation of the influence of small perturbation, as shown in Fig. 9d and e. Due to the asymmetric flow field, the central chimney moves towards left and merges with the central chimney. The enriched solute in the central chimney is transferred into the left chimney, promoting the stabilization of this chimney. In contrast, the right chimney is not as stable as the left one. The right chimney gradually weakens and disappears, while a channel develops continuously under the left chimney, as shown in Fig. 10e and f.

As mentioned before, channel segregation is found to distribute on the surface of specimen. In order to clarify the influence of melt flow in the near-wall region, another quantitative analysis of the flow field is carried out. The solute concentration and velocity on half of the lateral wall are shown in Fig. 10. Velocity is divided into tangential component and normal component, as shown in Fig. 8b. The tangential velocity component represents the local flow field on the surface, while the normal component reflects the melt flow from the centre. At 50 s, tangential velocity is almost symmetrical on both sides of the chimney, and is greater than the normal velocity, as shown in Fig. 10a. Obviously, chimneys driven by solute buoyancy dominate the local flow field at this time. At 200 s, normal velocity becomes much larger than tangential velocity, and the tangential velocity becomes asymmetric. More importantly, the peaks and valleys of the normal velocity and the solute distribution hardly overlap, which indicates that the melt convection from the centre will affect the solute field in the near-wall region, as shown in Fig. 9e and f. The channel segregation is enhanced when solute-enriched melt flows to the wall, while channel segregation is reduced when the solute concentration of the melt flowing to the wall is lower, as shown in Fig. 10b. As discussed before, the enrich solute will induce the upward flow and in turn change the local normal velocity, as shown in Fig. 10c.

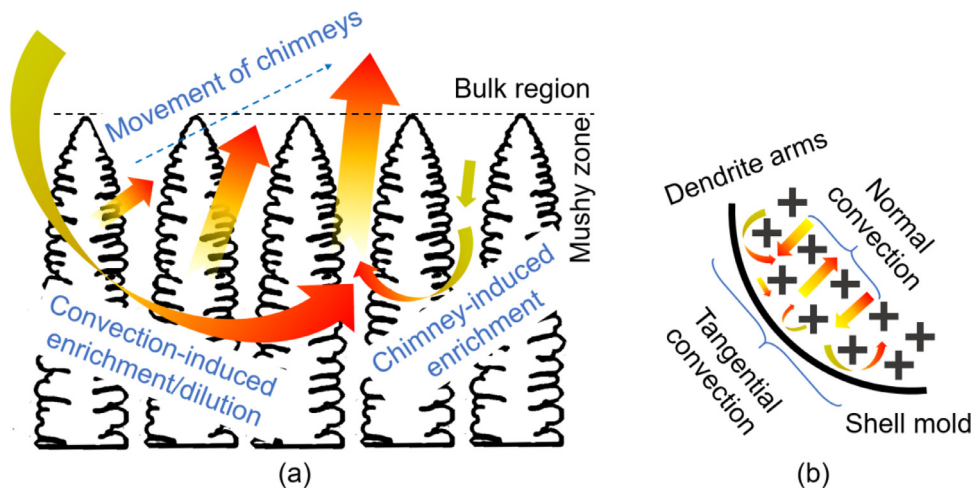


Fig. 8. Schematic of the interaction amongst solute distribution, melt convection and chimneys flow. (a) The mechanism of the movement and stabilization chimney flow. (b) Illustration of local normal and tangential convection in the near-wall region.

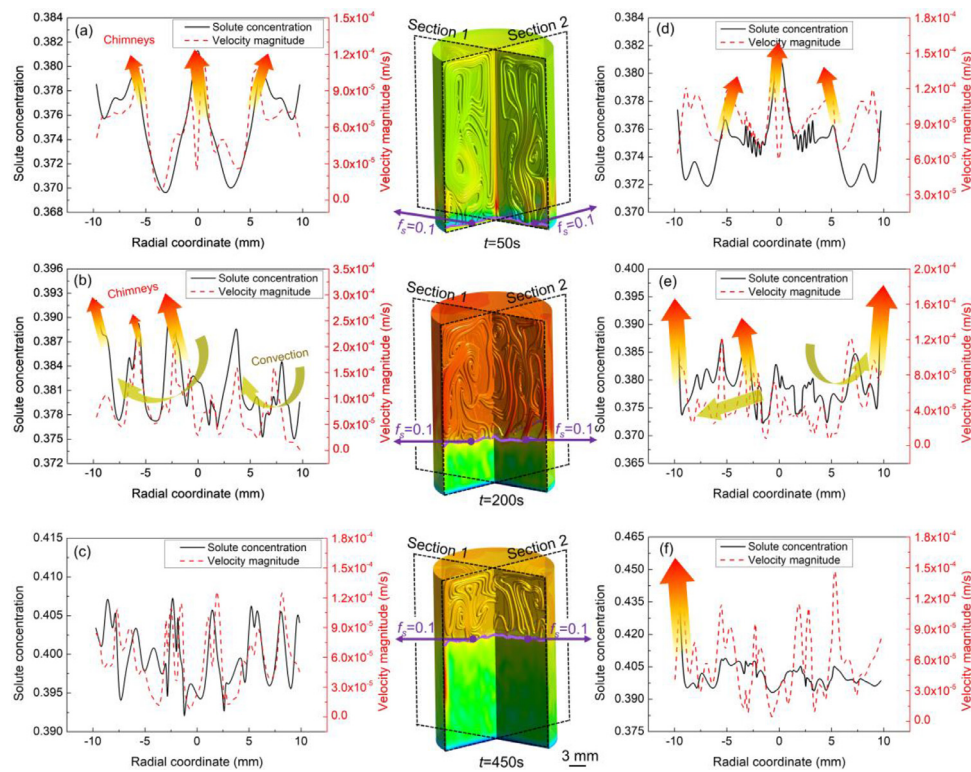


Fig. 9. The magnitude of horizontal velocity component and solute distribution along the solidification front on the longitudinal **Section 1** (a-c) and **Section 2** (d-f) at (a and d) 50 s, (b and e) 200 s and (c and f) 450 s. Iso-surface of $f_s=0.1$ (the purple line) is used to represent the solidification front. The sections and surfaces as well as the streamlines used to show the flow pattern are coloured in the magnitude of SI. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.2. How to control the freckles in single crystal blade

In order to further understand channel segregation in directional solidification, we perform two other cases with the convection coefficients of $20 \text{ W/m}^2\text{-K}$ and $100 \text{ W/m}^2\text{-K}$ to study the effect of lateral cooling condition on channel segregation. Fig. 11 shows the distribution of channel segregation in the specimens under different cooling conditions.

Compared with the benchmark case, no channel is observed when convection coefficient is $20 \text{ W/m}^2\text{-K}$, while more and shorter channels are formed when the convection coefficient is $100 \text{ W/m}^2\text{-K}$. At higher local solidification rates under higher lateral heat flux, as more solute is rejected into interdendritic region, chimneys are more likely to be formed and then induces solutal convection. The rejected solute from the surroundings is transported to the bottom of chimneys, promoting the formation of channel segregation. Thus, more channels are formed under higher lateral heat flux. However, the higher lateral cooling rate also induces a stronger descend flow driven by thermal buoyancy, leading to greater flow perturbation in the initial stage. As discussed above, it will cause the enriched solute to be transported along solidification front, thereby inhibiting the continuous development of channels at the bottom. As the solutal convection dominates the melt flow gradually, the perturbation effect of thermal convection declines, promoting effect starts to play a major role in channel developing. In short, the above results indicated that radial heat flux plays an important role that should not be ignored in the directional solidification of superalloys with Bridgman method, which is coincident with the previous investigations [47,56]. Reducing the lateral heat flux can be an approach in single-crystal nickel-based superalloys.

Another important difficulty in single-crystal blade casting is its complex geometry. In order to study the sensitivity of channel segregation to geometry, as well as to confirm the role of melt convection in the evolution of channel segregation, we conduct another case to simulate the channel segregation in quadrangular specimen. The evolution of channel segregation in a quadrangular specimen is shown in Fig. 12. Since heat loss at the corners is faster, stronger downward flow driven by thermal buoyancy suppresses the vortices near the corners, and thus the local solute is further concentrated at a higher growth rate. In the sharp features, because the local melt flow is less influenced by the surrounding flow field, chimneys can develop more stably. In contrast, the flow pattern in the region near lateral wall is more susceptible to the influence of surrounding melt convection, as shown in Fig. 12b and c. Therefore, a channel can extend from bottom to top at each corner, while only one channel with much lower segregation index is formed on lateral walls, as shown in Fig. 12d. In summary, higher cooling rate and weaker perturbation are the main reasons for the severe channel segregation in the sharp features. This is in accordance with the discussion in the previous sections.

In the previous investigations [22,56], the concave liquidus isotherm indicated the influence of radial heat flux was too large to be neglected in Bridgman method. As it is discussed above, channel segregation is less likely to be formed under a smaller lateral heat flux. In other words, it is possible to reduce and even eliminate freckles in blade castings by reducing the radial heat flux. Furthermore, the temperature field of a group of blades casting is asymmetric, lateral heat flux on the shadow side is higher. This means that freckles tend to appear on the shadow side of blade castings, as Ma et al. (2012) demonstrated in their experimental results [29]. According to the research on the effect of withdrawal rate [56,59],

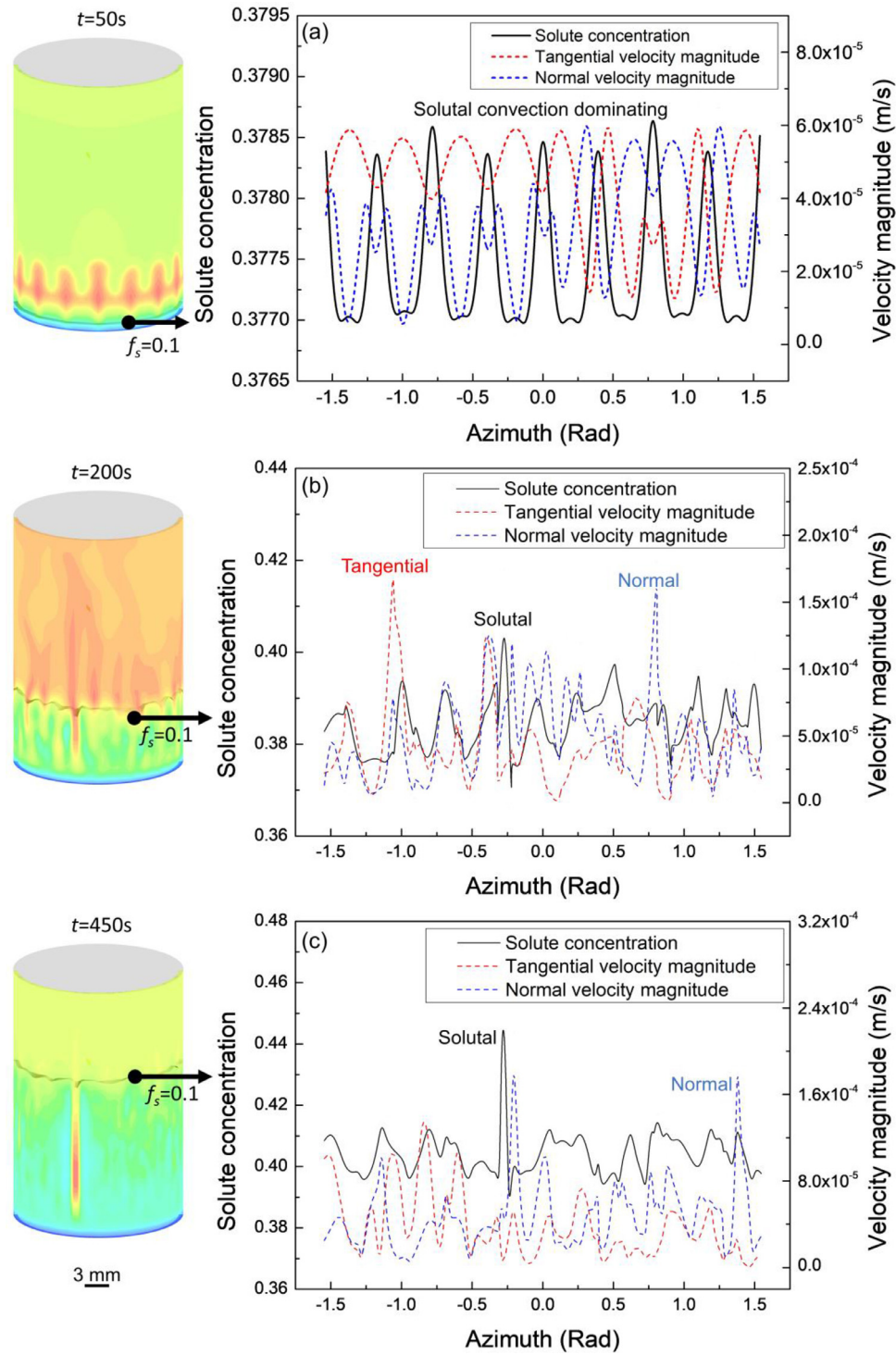


Fig. 10. Distribution of solute and velocity magnitude of normal and tangential component at (a) 50 s, (b) 200 s and (c) 450 s along the solidification front on the lateral wall. Iso-surface of $f_s = 0.1$ is used to represent solidification front. The data is extracted from the one layer of control volumes next to the lateral wall. The Cartesian coordinate is converted to cylinder coordinate here. Azimuth is used to show the position of control volumes on the lateral wall.

radial heat loss is enhanced if temperature gradient multiplied by withdrawal rate is higher than local cooling rate. Thus, in addition to carbon addition and increasing the temperature gradient, adjusting withdrawal rate can be another way to reduce freckles.

As discussed in the previous sections, the perturbation caused by melt convection inhibits the developing of channel segregation.

Based on this understanding, an external magnetic field and a coating on the interface between mould and melt can be employed to enhance the perturbation, thereby promoting the disappearance of the chimneys near the shell mould [60]. Due to the higher probability of channel segregation formation, sharp features should be avoided in the casting design if possible.

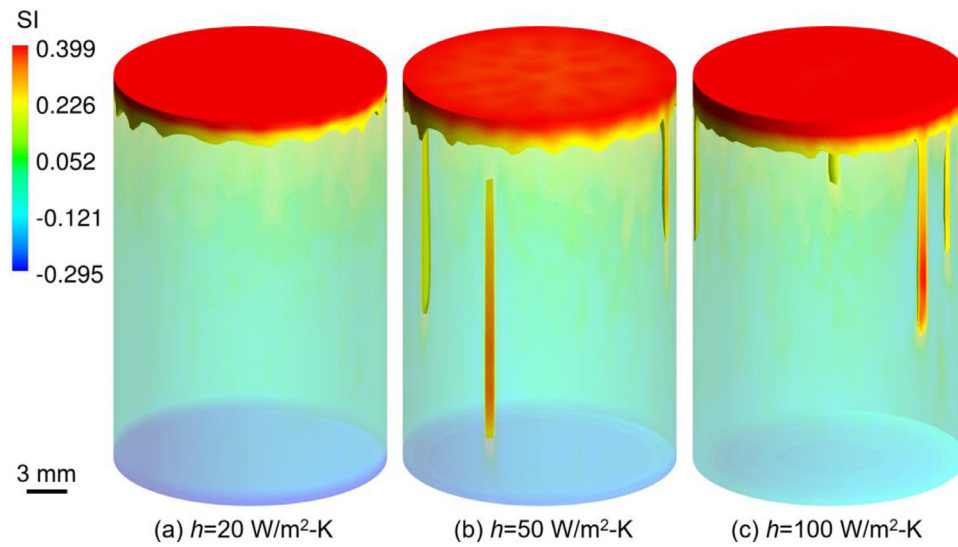


Fig. 11. The channel segregation distribution in the specimens under the convection coefficients of (a) 20 W/m²-K, (b) 50 W/m²-K and (c) 100 W/m²-K at 850 s.

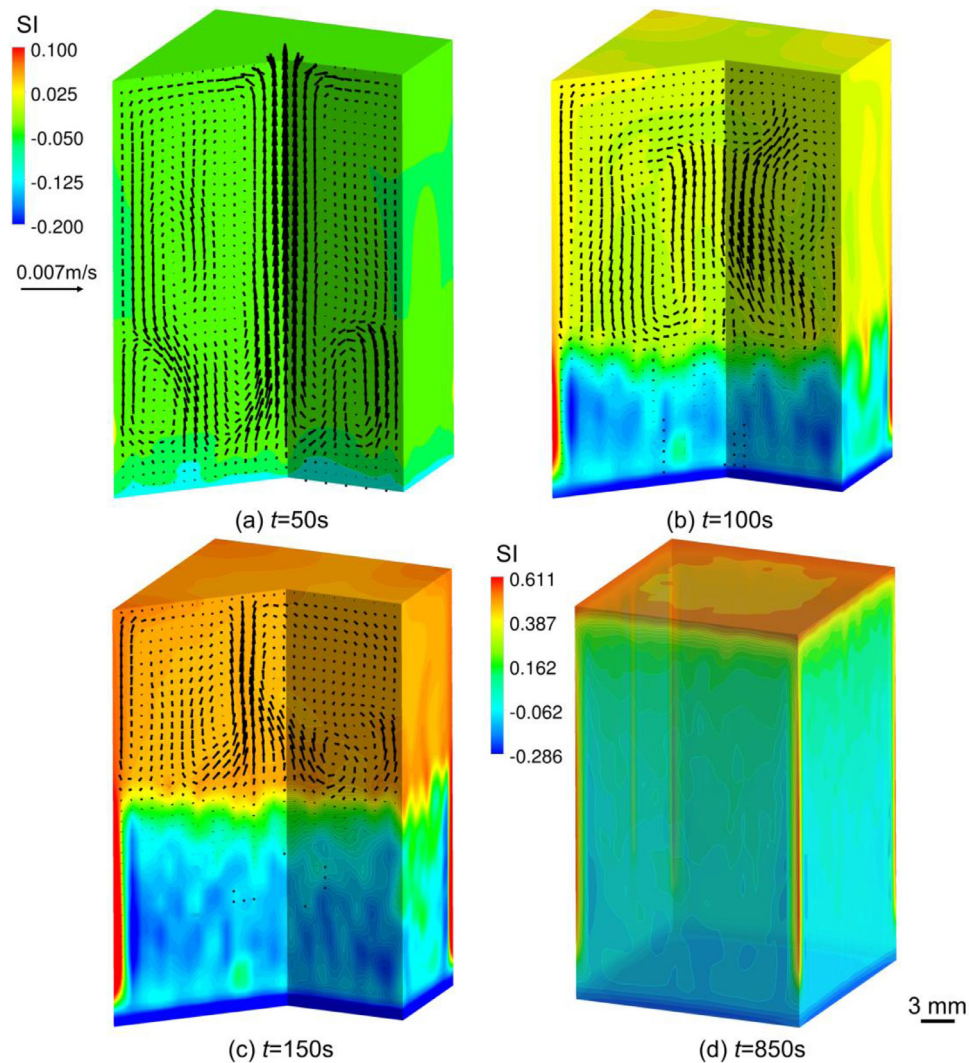


Fig. 12. Channel segregation distribution and flow pattern at (a) 50 s, (b) 100 s, (c) 150 s and (d) 850 s on the longitudinal sections in quadrangular specimen.

6. Conclusions

In the present work, we employ a three-dimensional Eulerian two-phase model to investigate the channel segregation of superalloys in directional solidification. The distribution and morphology of predicted channel segregation are in good agreement with experimental observations. The main conclusions are summarised as follows:

1. The heat flux on the lateral wall produces significant perturbation on the thermal-solutal convection of liquid metal in the directional solidification. Under the perturbation, not all the chimneys can induce channel segregation. Thus, the neglecting lateral heat flux in directional solidification leads to the inaccurate results when using Rayleigh number criterion to predict freckles.
2. Since more solute is rejected into liquid metal at a higher growth rate, chimneys are easier to form where the local solidification rate is higher. Under the influence of solutal convection, the surrounding rejected solute is transferred to the bottom of chimneys, leading to further solute enrichment and the stabilization of chimneys. Channels are therefore formed under the relatively stable chimneys. Additionally, in the initial stage when the density inversion in the mushy zone is not enough to induce chimney flow, downward flow driven by thermal buoyancy competes with upward driven by solutal buoyancy to inhibit the floating of the enriched solute, thereby promoting the initiation of channel segregation.
3. The perturbation of melt convection in mushy zone results in the disappearance of the channels. The channels in the centre are difficult to continuously develop upward under a sufficiently perturbation caused by unstable flow, while channels near the lateral wall are less affected and are more likely to develop stably along casting direction.
4. Due to the formation and disappearance of channels, channels are distributed in the thin layer of near-wall region. Similarly, channel segregation is also rather sensitive to cooling conditions and casting shape. The formation of chimney flow and channel segregation are promoted due to the solute enrichment in the near-wall region is enhanced under the higher heat flux. Though the perturbation is also enhanced under a higher heat flux, this perturbation attenuates as the stabilization of chimney flow, thereby just delaying the developing of channel. Thus, higher heat loss through the lateral wall leads to more shorter channels. Channels are easier to extend continuously in the sharp features where the local cooling rate is higher and solute distribution is less affected by unstable flow.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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